

Zhuoran (Lucy) Sun

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Engineer with 3+ years of experience delivering data/analytics solutions. Combines strong technical skills with business insight and internal client collaboration to drive fast, reliable, product-focused solutions for vaguely defined challenges

EDUCATION

Carnegie Mellon University

Master of Computational Data Science (Analytics Track)

GPA: **4.04/4.0**; Courses: Deep Reinforcement Learning, Machine Learning for Large Datasets, Cloud Computing

Pittsburgh, PA

Expected: Dec 2025

Emory University

Bachelor of Business Administration, Information Systems, Double Major: Math & CS

GPA: **3.95/4.0** (High Distinction, ISOM Distinction Award, Beta Gamma Sigma)

Atlanta, GA

May 2021

WORK EXPERIENCE

Keyfactor

Cleveland, OH/Atlanta, GA

Data Engineer Intern

June 2025 – Aug 2025

- Designed and led the implementation of data pipelines for Revenue Operations, automating data ingestion from Salesforce and ETL processes, which saved on average 5 hours of manual effort per product-level ARR report.
- Engineered and deployed an end-to-end application on Google Cloud Platform, collecting and cleaning geospatial data for thousands of US and EMEA leads. The system includes data collection, processing, and interactive dashboard to enable real-time geo-targeted sales analytics, cutting third-party licensing fees in thousands per month.
- Led a cross-functional project to develop a Minimum Viable Product for a Post-Quantum Cryptography awareness-scoring system for Fortune 500 companies. The system integrates web scraping to collect timely public news for the companies and performs NLP sentiment analysis to feed in the full-stack platform for scoring and visualization to communicate to non-technical stakeholders.

Truist Securities

Atlanta, GA

Quantitative Research Analyst/Associate

July 2021 – June 2024

- Led automation of daily PnL and risk reporting pipeline across fixed-income and loan trading, reducing generation time from hours to minutes with Pandas, NumPy, Snowflake and Tableau, and reporting directly to C-level managers. Developed a process to perform quarterly data scrutiny checks across multiple databases on 50K+ records.
- Delivered a POC analysis for a quartile-based bond trading strategy, utilizing statistical and time-series analysis, including ARIMA, Regression, on 3M+ request for quote data over 3+ years as part of the E-trading strategy initiative.
- Optimized RUNZ, an event-driven bond price listener using SpringBoot to capture Bloomberg quote messages, enabling price change alerts and pre and post trading time-series analysis for 200+ high-yield bonds concurrently.
- Took the initiative to perform SQL query optimization on the annual trading volume and balance sheet analysis for all trading businesses and reduced query generation time from 1 hour to 15 minutes for each of 6 trading desks.
- Team support: Built and distributed a Python API package on internal Artifactory to integrate pricing systems across derivative trading, ensuring stable ZMQ connectivity and secure compliance. Managed documentation and version control with Gitlab for all pipelines, processes, tools, and code in the team.

PROJECTS

Twitter Recommendation with Cloud Solutions, CMU

Pittsburgh, PA (Jan. 2025 – May 2025)

- Engineered a Spark ETL pipeline on 1TB+ Twitter data in Azure Databricks, designing an optimized data schema to calculate user recommendation scores and improve downstream micro-service performance
- Deployed production-level micro-services in Kubernetes clusters with Terraform, Docker, and Helm charts on AWS ECS, AWS EKS. Compared backend performance between MySQL, AWS RDS MySQL, and AWS ElastiCache service
- Fine-tuned the clusters to achieve latency within 5ms over a 3-hour live test and ranked top 5 in the leaderboard

Model Pruning for Efficient Deep Learning, CMU

Pittsburgh, PA (Mar. 2025 – May 2025)

- Performed pruning on a large CNN, achieving >94% sparsity while losing <1% accuracy, optimizing efficiency-accuracy tradeoffs for large-scale ML models, combining activation-based and network slimming methods

E-commerce Database Analysis, Emory University

Atlanta, GA (Aug. 2019 – July 2020)

- Analyzed 40M+ cart events from a major e-commerce platform using time-series models, random forests, LSTM, and DBSCAN in SciPy and PyTorch, boosting database efficiency and achieving 91% ROC in checkout prediction
- Paper accepted: OLTP In Real Life: A Large-scale Study of Database Behavior in Modern Online Retail at MASCOTS 2021

Sales System, The Emory Wheel

Atlanta, GA (Sept. 2017 – June 2020)

- Migrated sales data from Google Sheets to a Django/PostgreSQL system on Heroku, enabling reporting, analytics, and commission tracking, saving the sales manager 50% of time